

A finite-size active advantage over number-sector-incoherent passive sensing for four-fold forrelation at $N = 4096$

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We consider a four-block binary-phase sensing task with $N = 4096$ modes per block, where every photon traversal of an unknown phase coordinate is charged, including traversals preceding discarded or no-click outcomes. The task distinguishes four-fold forrelation at least $1/4$ from four-fold forrelation at most $-1/4$. We prove that every classically adaptive passive protocol of hard dose at most six has worst-case promised error greater than $1/3$ when each fresh batch is block diagonal in total signal photon number. The class permits arbitrary idlers, repeated modes, within-sector entanglement and coherence, collective measurements, and unrestricted classical feed-forward, but no coherent quantum memory between batches. The passive proof combines an attenuated signed-permutation hard pair, 888 physical occurrence bounds, an outward 210-state spectral certificate, and a normalized adaptive factorization, giving transcript distance at most 0.2609692247922079. Coherence between different signal-number sectors, including vacuum–nonvacuum coherence, is not covered. An active protocol retains one photon coherently across two charged phase grids; three independent flags and majority decoding use hard dose six and have worst-case error $81/256$. The ideal protocol requires 16,384 sign coordinates and retained contrast above 0.904294855157, and current platform evidence does not certify an implementation at this scale.

I. INTRODUCTION

Dose-limited sensing assigns a physical cost to each interaction between a probe and a sample, because absorption, scattering, photochemistry, heating, and structural change can make repeated interrogation more expensive than source preparation, receiver complexity, or classical computation. The comparison studied here uses the branchwise hard photon-pass dose, under which every photon crossing an unknown sample coordinate contributes one unit on the realized branch, without discounts for heralding, postselection, erasure, or a detector outcome that is later ignored.

The sample consists of four binary phase masks $x^{(1)}, \dots, x^{(4)} \in \{\pm 1\}^N$, each acting on N optical modes, with fixed normalized Sylvester transforms between the masks. The measured quantity is the four-fold forrelation amplitude, which is the coherent return amplitude of a uniform mode after the alternating mask and transform chain. Forrelation was introduced as a query-complexity separation and subsequently developed into a family of Fourier-correlation problems with strong classical lower bounds [1, 2]. The present use treats the sign strings as physical phase patterns and compares two quantum access classes under the same sample-interaction meter.

A passive protocol prepares a fresh quantum batch, applies the unknown sign oracle once to the batch, performs an arbitrary collective measurement, stores the classical outcome, and chooses the next fresh batch from the accumulated record. The certified passive class permits arbitrary mixtures of total signal photon numbers, but each batch state must commute with the total signal photon-number operator; it may otherwise contain arbitrary within-sector entanglement and coherence, idlers, and repeated physical modes. An active protocol may retain a live quantum system between charged sample

interactions and may apply known coherent transformations before returning to the sample. The additional restriction on passive number-sector coherence is mathematically substantive and is not implied by the absence of inter-batch quantum memory.

A provisional asymptotic companion derivation proves a passive hard-dose lower bound $D \geq (2/15)N^{1/8}$ for powers of two $N \geq 2^{30}$ in the full fresh-probe model, together with an active upper bound of six [3]. The present paper establishes the finite instance $N = 4096$ for the number-sector-incoherent fresh-batch subclass, where the asymptotic theorem does not determine whether dose six is excluded. The finite proof requires a separate coefficient registry, exact physical distinctness masks, a numerical spectral certificate with directed rounding, and an adaptive factorization whose normalization remains one for arbitrarily wide and deep classical outcome trees.

The principal result is

$$D_{\text{fresh}}^{\text{hard}}(4096) > 6, \quad D_{\text{A}}^{\text{hard}}(4096) \leq 6, \quad (1)$$

for worst-case promised error at most $1/3$. For two transcript laws, total variation distance is the largest difference they assign to the same decision event. The passive proof gives the outward bound

$$\text{TV}(P_+, P_-) \leq 0.2609692248, \quad (2)$$

which implies average error at least 0.3695153876 under the equal-prior hard pair; Sec. III gives the complete directed decimals. The active protocol returns three binary flags whose individual expectation equals the forrelation amplitude; majority decoding has worst-case error $81/256$ at the promise boundary and uses two charged traversals per flag.

The proof has four components. First, three independent signed permutations of order $q = 64$ generate an exact four-block plant with forrelation $+1$, and independent biased signs with mean $\beta = 19/25$ attenuate the plant before conditioning it onto the two promise sides. Second, every balanced degree-four through degree-twelve physical occurrence matrix relevant to the number-sector-incoherent class receives an arbitrary-correlated-diagonal coefficient, with all cross-cut distinctness masks retained, producing a complete high-sector registry of 888 entries. Third, the coefficients assemble into a nonnegative symmetric matrix on the 210 four-block occupation vectors of total dose at most six, block diagonal in total occupation, whose Perron root is bounded by an outward Collatz–Wielandt certificate. Fourth, a normalized direct-sum feature factorization combines every outcome-selected continuation of a passive tree into row and column masses at most one, so the one-batch occurrence coefficients and the 210-state spectral bound apply without an adaptive multiplier.

The active circuit uses four sign banks, $M = 4N = 16,384$ unknown sign coordinates, a coherent logical mode space of dimension 4096, one path qubit, and three single photons. The ideal majority decoder requires combined zero-bias retained contrast greater than 0.904294855157. Published detector, time-bin, and spatial-mode demonstrations provide relevant component evidence [4–7]. None of these cited demonstrations reports a complete 4096-dimensional coherent Sylvester implementation at the required end-to-end contrast, so the implementation assessment remains separate from the mathematical separation.

Section II defines the task, access classes, and meter. Section III states the theorem and numerical certificate. Sections IV–VII give the active protocol and the passive lower bound. Section VIII states the implementation requirements, Sec. IX records the interpretation and limitations, and Sec. X specifies the reproducibility package.

II. THE MODEL

A. Definition

Let $N = 4096$, let $H = H_N$ be the normalized Sylvester matrix, and let $x = (x^{(1)}, x^{(2)}, x^{(3)}, x^{(4)})$ with $x^{(j)} \in \{\pm 1\}^N$. The diagonal sign operation associated with block j is

$$D_j = \text{diag}(x^{(j)}). \quad (3)$$

Each sign is a binary phase shift on one physical mode, with $+1$ and -1 corresponding to relative phases zero and π . The normalized Sylvester matrix is a fixed balanced transformation that distributes one input amplitude uniformly over all output modes with phases in $\{0, \pi\}$.

The four-fold forrelation amplitude is

$$\begin{aligned} F_{4,H}(x) &= \langle u | D_1 H D_2 H D_3 H D_4 | u \rangle \\ &= \frac{1}{N} (x^{(1)})^T H D_2 H D_3 H x^{(4)}, \end{aligned} \quad (4)$$

where $|u\rangle = N^{-1/2} \sum_i |i\rangle$. The physical reading of Eq. (4) is a coherent return amplitude: a uniform single-photon mode acquires four phase patterns separated by three balanced transformations, and the final amplitude is projected back onto the uniform mode.

Definition 1 (Promise). The positive promise set contains every x with $F_{4,H}(x) \geq 1/4$, and the negative promise set contains every x with $F_{4,H}(x) \leq -1/4$. A protocol succeeds at error ϵ when its error probability is at most ϵ for every input in both promise sets.

B. The meter

A charged interaction is one photon pass through one unknown sign coordinate. A multi-photon Fock component with total occupation t across the four sign banks has charge t for one simultaneous oracle application. If a coherent routing operation places amplitude on paths with different numbers of charged interactions, the branch charge is the maximum supported number because the protocol must be physically capable of the longest path while coherence is retained.

Definition 2 (Branchwise hard dose). For an adaptive protocol represented as an outcome tree, the hard dose is the maximum total number of photon passes over all root-to-leaf branches. Every charged interaction is included, including interactions preceding no-click outcomes, erased records, rejected heralds, or later postselection.

The hard meter assigns the same cost to the passive and active classes. Known transforms, lossless routing, delay, idler storage, and measurements do not contribute sample dose, although they remain physical resources in the implementation statement.

C. Terms of the comparison

Definition 3 (Number-sector-incoherent classically adaptive passive access). A protocol in $\mathcal{P}_{\text{fresh}}^{\text{ns}}$ is a finite-depth outcome tree. At each node it prepares a fresh signal–idler state ρ satisfying

$$[\rho, \hat{N}_{\text{sig}} \otimes I_{\text{id}}] = 0, \quad (5)$$

applies the four-bank diagonal sign oracle once collectively to the signal, performs an arbitrary collective POVM on the signal and idlers, records the classical outcome, and selects the next fresh batch from the complete classical history. The model permits arbitrary mixtures of total signal

photon numbers, arbitrary coherence and entanglement within each fixed-number sector, idlers, repeated physical modes, mixed probes, randomized controllers, general measurable outcome spaces, and branch-dependent dose partitions. It excludes coherence between different total signal-number sectors, including vacuum–nonvacuum coherence. No coherent quantum system or coherent control is carried from one batch to the next.

The term *fresh* refers to the causal boundary between charged interactions: after a batch measurement, all information available to later nodes is classical. The number-sector condition is a separate restriction. The theorem does not impose product laws on the row or column weights that arise in the proof, because outcome selection can correlate complete temporal histories.

Definition 4 (Active access). A protocol in **A** may retain a coherent quantum system across multiple charged sign-oracle applications and may apply arbitrary known unitaries, routing operations, and delays between them. Every sign-oracle application remains charged under the branchwise hard meter.

The active protocol in Sec. IV uses this permission only to maintain one photon coherently across two sign-bank traversals. The theorem makes no statement about passive protocols augmented with coherent quantum memory between fresh batches, because that augmentation changes the access class.

D. Decision complexity

Let $D_C^{\text{hard}}(N)$ be the minimum branchwise hard dose of a protocol in class **C** whose worst-case promised error is at most $1/3$. The finite-size comparison concerns $D_{\text{P}_{\text{fresh}}^{\text{ns}}}^{\text{hard}}(4096)$ and $D_A^{\text{hard}}(4096)$ under the definitions above. The corresponding lower bound for fresh passive probes with coherence between different signal-number sectors remains open.

III. THE FINITE-SIZE STATEMENT

Theorem 1 (Finite-size number-sector-incoherent passive separation). *For the four-fold forrelation promise problem at $N = 4096$, every protocol in $\text{P}_{\text{fresh}}^{\text{ns}}$ with branchwise hard dose at most six has worst-case promised error greater than $1/3$, while a protocol in **A** with branchwise hard dose six has worst-case promised error $81/256$. Therefore*

$$\boxed{D_{\text{P}_{\text{fresh}}^{\text{ns}}}^{\text{hard}}(4096) > 6, \quad D_A^{\text{hard}}(4096) \leq 6.} \quad (6)$$

The passive lower bound covers unrestricted classical feed-forward between fresh quantum batches satisfying Eq. (5). It excludes coherent quantum memory or coherent control carried between batches and coherence between different total signal photon-number sectors within a batch.

a. Claim grade. Theorem 1 is theorem-level. Its passive numerical input is an outward certificate, its adaptive extension is analytic with multiplier one, and its active upper bound is pointwise on every promised input.

A. Numerical certificate

The passive proof uses two conditioned hard laws P_+ and P_- . The following displayed upper bounds are rounded upward from the exact directed-arithmetic artifacts.

$$P_{\text{Perron}} \leq 0.258744096385577223, \quad (7)$$

$$P_{\text{promise}} \leq 0.002225128406630703, \quad (8)$$

$$\text{TV}(P_+, P_-) \leq 0.260969224792207925. \quad (9)$$

The acceptance threshold used by the certificate is

$$\frac{1}{3} - 10^{-3} = 0.33233333333333333333 \dots, \quad (10)$$

so the outward total retains the lower reserve

$$0.071364108541125408. \quad (11)$$

For equal priors, the Bayes error is $(1 - \text{TV})/2$, and Eq. (9) gives

$$p_{\text{err}}^{\text{passive}} \geq 0.369515387603896037 > \frac{1}{3}. \quad (12)$$

An average error lower bound over hard distributions supported on the two promise sets implies that every passive hard-dose-six protocol fails on at least one promised input.

B. Reading the separation

At dose six, the active error $81/256 = 0.31640625$ lies $13/768 \simeq 0.01693$ below the required error $1/3$. The passive hard-pair error is at least 0.369515387603896037 , which lies approximately 0.03618 above $1/3$; because this is an average over promised inputs for an arbitrary protocol in $\text{P}_{\text{fresh}}^{\text{ns}}$, at least one promised input has error no smaller than this average. Figure 1 displays the certificate assembly and the two resulting decision-error statements on a common scale.

C. Dependency structure

The proof dependencies are ordered. The physical occurrence theorems establish arbitrary-correlated-diagonal

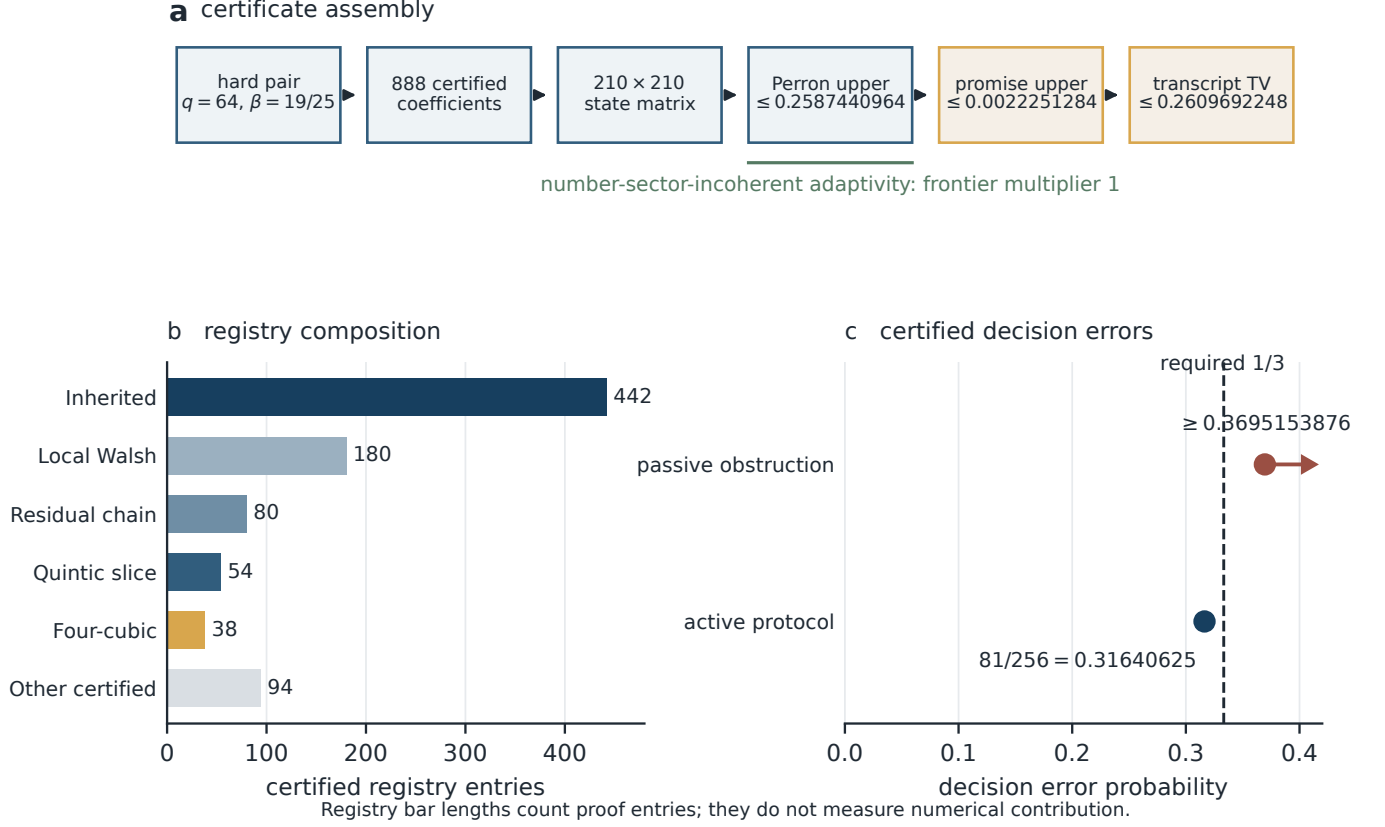


FIG. 1. Assembly of the number-sector-incoherent passive certificate and the resulting finite-dose separation. The 888 balanced arbitrary-correlated-law occurrence coefficients assemble the 210-state matrix, whose outward Collatz–Wielandt upper is 0.2587440964; the normalized adaptive factorization contributes multiplier one, and promise conditioning contributes at most 0.0022251285. The passive equal-prior hard-law error is at least 0.3695153876, which implies a worst-case promised obstruction in the stated passive class, while the explicit active protocol has pointwise worst-case error 81/256. Registry bar lengths count certified entries and do not measure their numerical contributions.

coefficients for all balanced entries. The one-batch program rejects every unbalanced routing value, replaces each accepted coefficient by a strict rational upper, reconstructs the number-sector-block-diagonal 210-state nonnegative matrix with directed arithmetic, and certifies its spectral radius. The adaptive theorem factors the complete outcome tree into one normalized feature family, pulls temporal histories back to aggregate Fock occupations, and applies the same occurrence matrix to two independent normalized laws. Promise conditioning is paid once after the unconditioned comparison. The active upper bound uses a separate pointwise interferometric identity and does not depend on the passive hard distribution.

IV. THE ACTIVE PROTOCOL

A. Folded states

Define

$$|L_x\rangle = D_2 H D_1 |u\rangle, \quad |R_x\rangle = H D_3 H D_4 |u\rangle. \quad (13)$$

One photon is prepared in

$$\frac{|0\rangle|L_x\rangle + |1\rangle|R_x\rangle}{\sqrt{2}}, \quad (14)$$

where the first register is a path degree of freedom and the second register contains 4096 logical modes. Chronologically, the left path crosses D_1 , then H , then D_2 , while the right path crosses D_4 , then H , then D_3 , then a second H . Both paths therefore cross exactly two unknown sign banks.

The folded-state overlap is

$$\langle L_x | R_x \rangle = \langle u | D_1 H D_2 H D_3 H D_4 | u \rangle = F_{4,H}(x). \quad (15)$$

Figure 2 gives the physical circuit for one flag. Every unknown sign bank is shown as a two-dimensional phase

TABLE I. Active resource ledger at $N = 4096$.

Resource	Requirement
Sign banks	4
Sign coordinates	16,384
Logical mode dimension	4096
Path dimension	2
Single-photon flags	3
Charged traversals per flag	2
Total hard dose	6
Receiver	path- X , mode insensitive
Postselection	none

grid indexed by $(a, b) \in [64]^2$; this coordinate display follows the factorization $4096 = 64^2$ and does not require the device to use a square spatial array.

A Pauli- X measurement on the path produces a binary flag $X \in \{\pm 1\}$ with

$$\mathbb{E}[X | x] = F_{4,H}(x). \quad (16)$$

The mode register need not be resolved at the receiver; the two paths are recombined, the output port is detected, and the mode label is summed over.

B. Error and dose

At either promise boundary, one flag is correct with probability $5/8$. Three independent flags followed by majority decoding have worst-case error

$$\left(\frac{3}{8}\right)^3 + 3\left(\frac{5}{8}\right)\left(\frac{3}{8}\right)^2 = \frac{81}{256} = 0.31640625. \quad (17)$$

The exact margin below $1/3$ is

$$\frac{1}{3} - \frac{81}{256} = \frac{13}{768}. \quad (18)$$

For example, the retained detector record $(+1, -1, +1)$ is decoded as the positive promise side. Every one of the three flags remains in the record, so the majority rule uses no favorable-outcome filtering or postselection.

Each flag uses deterministic hard dose two, and the three retained flags use deterministic hard dose six. The protocol has no postselection, and all no-click or erased outcomes must remain in the decision record under a nonideal implementation.

The three flags may use the same optical network sequentially, so the resource statement does not require three copies of the sign banks or public transforms.

V. THE HARD PAIR

A. Signed-permutation plants

Set $q = 64$, so $N = q^2$, and write $S_q = \sqrt{q} H_q$ for the unnormalized Sylvester sign matrix. Coordinates in each length- N block are ordered by row-major flattening of a $q \times q$ array, under which $H_N = H_q \otimes H_q$.

A signed permutation is specified by a uniform permutation π of $[q]$ and independent uniform signs σ_b . Associate two sign arrays

$$L_{a,b} = S_q(a, \pi(b))\sigma_b, \quad R_{\pi(b),a} = \sigma_b S_q(b, a). \quad (19)$$

For three independent signed permutations P_1, P_2, P_3 , define the four positive-plant blocks

$$z^{(1)} = L(P_1), \quad z^{(2)} = R(P_1) \odot L(P_2), \quad (20)$$

$$z^{(3)} = R(P_2) \odot L(P_3), \quad z^{(4)} = R(P_3), \quad (21)$$

after flattening. The signed-permutation identities telescope through the three Sylvester transforms and give

$$F_{4,H}(z) = 1. \quad (22)$$

Replacing $z^{(1)}$ by $-z^{(1)}$ gives an exact negative plant with value -1 .

Each array entry in Eq. (21) is one binary phase coordinate, with $+1$ representing phase zero and -1 representing phase π . The shared $R(P_j)$ and $L(P_{j+1})$ factors correlate adjacent grids so that their signed-permutation indices contract through the intervening Sylvester transform. Figure 3 shows one deterministic realization at the theorem value $q = 64$.

B. Attenuation

Fix the rational attenuation

$$\beta = \frac{19}{25} = 0.76. \quad (23)$$

Every coordinate in every plant block is multiplied independently by a sign η with

$$\Pr(\eta = 1) = \frac{1 + \beta}{2} = 0.88, \quad (24)$$

$$\Pr(\eta = -1) = \frac{1 - \beta}{2} = 0.12. \quad (25)$$

The attenuated positive plant has mean correlation β^4 , while reflection of the first block gives the negative counterpart. Attenuation spreads each exact orbit over many sign strings and reduces the low-degree distinguishability available to a dose-six passive protocol.

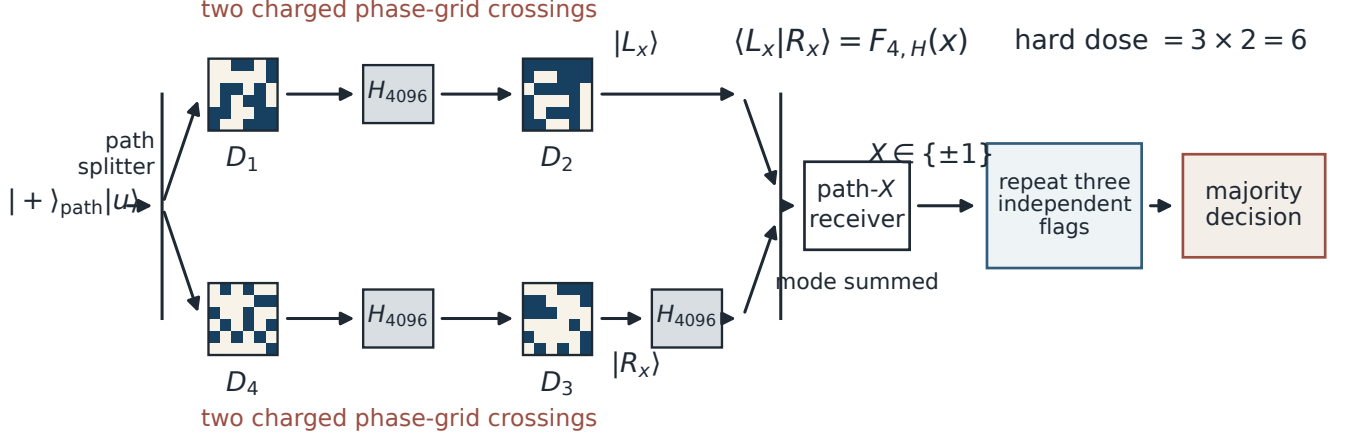


FIG. 2. One active binary flag and the three-flag decision. The left path crosses D_1 , H_{4096} , and D_2 , while the right path crosses D_4 , H_{4096} , D_3 , and a second H_{4096} . Each path therefore crosses two charged phase grids. Recombination in the path- X basis measures the folded-state overlap while summing over the logical mode coordinate. Three sequential repetitions use the same four sign banks, have total hard dose six, and are decoded by majority. The phase patterns are representative two-dimensional grid icons rather than a restriction on the input law.

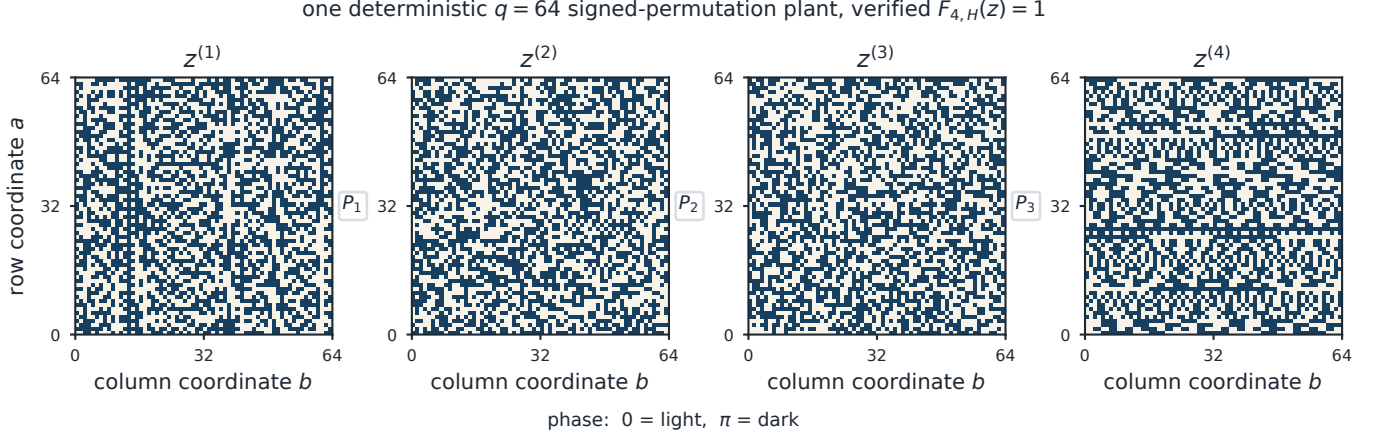


FIG. 3. One deterministic positive signed-permutation plant at $q = 64$. The four panels display the 64×64 coordinate arrays $z^{(1)}, \dots, z^{(4)}$, with the two colors denoting phases zero and π ; vector enlargement resolves the individual cells. Adjacent blocks share the indicated signed permutation through their $L(P_j)$ and $R(P_j)$ factors, and an exact integer Walsh–Hadamard contraction verifies $F_{4,H}(z) = 1$ for the displayed realization. The square grid represents the logical coordinate pair (a, b) and is not a claim about physical layout.

C. Conditioning

Let $\tilde{P}_+$ and $\tilde{P}_-$ be the two unconditioned attenuated laws. The hard laws used in the theorem are

$$P_+ = \tilde{P}_+(\cdot \mid F_{4,H} \geq 1/4), \quad (26)$$

$$P_- = \tilde{P}_-(\cdot \mid F_{4,H} \leq -1/4). \quad (27)$$

The two laws are therefore supported entirely on the two promise sets. The hard pair is a lower-bound witness and is not part of the definition of the task or the active protocol.

The accepted concentration argument bounds the sum of the two bad-promise probabilities by

$$P_{\text{promise}} \leq 0.002225128406630703. \quad (28)$$

This cost is added once after the unconditioned transcript comparison. Adaptive posterior nodes are not separately conditioned.

VI. THE ONE-BATCH CERTIFICATE

A. Occurrence expansion

A dose-six batch is expanded in Fock occupations on the four sign banks. When a ket occupation and a bra occupation are compared under the hard pair, their symmetric difference produces a profile $a = (a_1, a_2, a_3, a_4)$ of total degree between four and twelve, and a split t records which occurrences lie on one side of the matrix. Exact occupation compatibility is

$$m = n + a - 2t, \quad (29)$$

where n and m are the four-block ket and bra occupations. The state space contains every occupation with total dose at most six. Because Eq. (5) removes ket–bra terms of different total signal photon number, a contributing profile–split pair must satisfy $2|t| = |a|$. Equation (29) then preserves total occupation.

The upstream routing inventory contains 6,016 high-sector profile–split records. Exactly 888 satisfy the number–sector balance condition, and every one of those records has a proved coefficient. The remaining 5,128 routing values are excluded from the theorem ledger rather than called coefficient bounds. This distinction matters: 272 of those unbalanced profile splits would create split/state incidences, comprising 136 undirected edges between different-total-number states, if coherence between number sectors were allowed. Their physical coefficients are not certified by this package.

As a concrete occupation example, take

$$n = (1, 1, 1, 1), \quad a = (1, 1, 1, 1), \quad t = (1, 0, 1, 0). \quad (30)$$

Equation (29) gives $m = (0, 2, 0, 2)$, so both sides have dose four because $2|t| = |a| = 4$. The four coordinates in these vectors are photon totals in the four sign banks rather than positions within a bank; the physical positions and their distinctness masks remain inside the occurrence matrix $\mathcal{O}_{a,t}$.

B. Physical coefficients

For a fixed profile and split, the physical occurrence matrix retains the within-block simple-support constraints and every cross-cut distinctness mask. Its accepted coefficient $c_{a,t}$ bounds

$$\sup_{p,w} \|\text{diag}(\sqrt{p}) \mathcal{O}_{a,t} \text{diag}(\sqrt{w})\|_1 \leq c_{a,t}, \quad (31)$$

for arbitrary correlated diagonal row and column laws p, w . Product laws, uniformity, exchangeability, translation invariance, and separate optimization of different links are not assumed.

The physical coefficient proofs are grouped into masked quintic slices, local Walsh contractions, cubic and quintic endpoint factorizations, finite incidence theorems, shared-quintic contractions, a final residual-chain theorem, and an independently audited dual-endpoint theorem. Earlier coefficient-one arguments that omitted cross-cut masks are absent from the accepted registry. Appendix B gives the status counts and source map.

C. Outward coefficient rounding

Every accepted nonnegative binary64 coefficient c is replaced by

$$\hat{c} = \frac{\lceil 10^9 c \rceil + 1}{10^9} > c. \quad (32)$$

The extra grid unit makes the replacement strict and limits the inflation to less than 2×10^{-9} . Coefficients supplied as exact rational squares are checked separately against the squares of their outward ledger values.

D. Occupation matrix

There are 210 four-block nonnegative occupation vectors of total dose at most six. The rounded coefficients and attenuation weights assemble into a symmetric entrywise nonnegative matrix B on these states. The occurrence contribution for normalized occupation laws is bounded by the spectral norm of B , which equals its Perron root.

A floating eigensolve of the balanced matrix supplies a strictly positive candidate vector v and does not supply the accepted eigenvalue. The certified value uses the Collatz–Wielandt inequality

$$\rho(B) \leq \max_i \frac{(\hat{B}v)_i}{v_i}, \quad (33)$$

where $\hat{B}$ is reconstructed entrywise with 80-digit directed arithmetic and outward square roots. This gives

$$P_{\text{Perron}} \leq 0.258744096385577223. \quad (34)$$

Adding Eq. (28) gives the one-batch outward total in Eq. (9). The remaining question is whether arbitrary classical outcome trees whose fresh probes obey Eq. (5) preserve the coefficient and spectral normalizations; Sec. VII supplies that lift.

VII. THE ADAPTIVE LIFT

A. Strategy factorization

Fix terminal weights in $[-1, 1]$ for the signed transcript functional of a finite passive tree. A complete temporal

ket history is denoted by I , a complete temporal bra history by J , and the signed strategy coefficient after removing the oracle characters by $D(I, J)$.

Outcome-dependent fresh probes can correlate complete histories because a later preparation may depend on every earlier measurement result. By convexity, a state satisfying Eq. (5) may first be refined into a classical mixture of pure probes, each supported in one fixed total signal-number sector; revealing that mixture label to the controller can only enlarge the passive class. The following factorization retains the resulting correlations while showing that classical outcome selection cannot create additional normalized interference mass: every mixture label and child is placed in an orthogonal direct-sum sector, and normalization and POVM completeness bound the total squared feature mass by one.

Lemma 1 (Normalized strategy factorization). *There are Hilbert-space features u_I, v_J such that*

$$D(I, J) = \langle v_J, u_I \rangle, \quad (35)$$

$$\sum_I \|u_I\|^2 \leq 1, \quad \sum_J \|v_J\|^2 \leq 1. \quad (36)$$

The statement holds for arbitrary outcome width, depth, rare outcomes, outcome-selected number-sector-incoherent child probes, and branch-dependent dose partitions.

Conditional on its refined classical mixture label, write the first fresh probe as

$$|\psi\rangle = \sum_i |i\rangle |\alpha_i\rangle, \quad \sum_i \|\alpha_i\|^2 = 1, \quad (37)$$

where every occupied basis label i in this sum has the same total signal photon number. Taking one additional direct sum over the classical mixture labels preserves the normalization below.

and let $\{E_y\}_y$ be the root POVM. Define

$$a_{y,i} = E_y^{1/2}(|i\rangle |\alpha_i\rangle). \quad (38)$$

POVM completeness gives $\sum_{y,i} \|a_{y,i}\|^2 = 1$. If the child selected by outcome y has features $u_{y,I}, v_{y,J}$, define

$$U_{i,I} = \bigoplus_y a_{y,i} \otimes u_{y,I}, \quad V_{j,J} = \bigoplus_y a_{y,j} \otimes v_{y,J}. \quad (39)$$

The direct sum reproduces the complete signed coefficient and satisfies

$$\sum_{i,I} \|U_{i,I}\|^2 \leq \sum_{y,i} \|a_{y,i}\|^2 = 1, \quad (40)$$

with the same bound for the column features. Rare outcomes introduce no inverse probability because the proof does not normalize posterior laws.

B. Temporal occupation pullback

For a temporal history $I = (i_1, \dots, i_d)$, define its aggregate physical Fock occupation

$$f(I) = i_1 + \dots + i_d. \quad (41)$$

Diagonal sign characters multiply according to

$$\chi_{i_1}(x) \cdots \chi_{i_d}(x) = \chi_{f(I)}(x). \quad (42)$$

Repeated physical modes remain in the integer occupation and reduce through parity in the character. Orthogonality of the refined number-sector labels makes the strategy cross-Gram symbol zero unless ket and bra histories have equal total signal photon number in every batch, and hence $|f(I)| = |f(J)|$ in every contributing term. The branchwise hard-dose condition gives $|f(I)| \leq 6$. Every adaptive hard-instance moment is therefore a pullback of the same balanced degree-four through degree-twelve physical kernels certified in Sec. VI.

Duplicating a physical row or column through several temporal decompositions does not increase a Hilbert factorization norm. Setting $p_I = \|U_I\|^2$ and $w_J = \|V_J\|^2$ produces two generally correlated laws with total mass at most one, which are within the arbitrary-law scope of Eq. (31).

C. The Perron interface

Push p and w forward to the four-block occupations of $f(I)$ and $f(J)$. For the same nonnegative symmetric 210-state matrix B , the complete occurrence sum has the form $x^T B y$, where $\|x\|_2, \|y\|_2 \leq 1$. Hence

$$x^T B y \leq \|B\|_{2 \rightarrow 2} = \rho(B). \quad (43)$$

The outward Collatz-Wielandt certificate therefore accepts the two independent laws generated by the strategy features within each total-number block. The adaptive occurrence multiplier is exactly one.

Promise conditioning is applied once to the completed unconditioned transcript comparison, giving the same total as Eq. (9). Lemma 1 covers classical feed-forward between number-sector-incoherent fresh batches. It does not extend to coherent quantum systems retained between batches or to fresh probes with coherence between different total signal-number sectors.

VIII. PHYSICAL IMPLEMENTATION

A. Required components

The active protocol requires a heralded or deterministic single photon in a coherent uniform superposition over

4096 logical modes, a binary path degree of freedom, four independently specified 4096-coordinate sign banks, one coherent H_{4096} transformation on the left path, two coherent H_{4096} transformations on the right path, stable relative phase between the paths, and a mode-insensitive two-port receiver. Three photons may traverse the same apparatus sequentially.

The number 16,384 counts unknown sign coordinates across four reusable banks. It is not a detector-pixel requirement. The receiver resolves only the binary output port while summing over the logical mode register.

B. Contrast requirement

Let the retained signed expectation of one binary flag at the promise boundary be μ . Three independent flags have majority error

$$\epsilon_3(\mu) = \left(\frac{1-\mu}{2}\right)^2 (2+\mu). \quad (44)$$

The equation $\epsilon_3(\mu_*) = 1/3$ has the relevant root

$$\mu_* = 0.226073713789207990\dots \quad (45)$$

Since the ideal boundary expectation is $1/4$, the zero-bias multiplicative contrast must satisfy

$$g > g_* = 4\mu_* = 0.904294855156831964\dots \quad (46)$$

In the scalar allocation

$$g = \xi \sqrt{T_L T_R} \nu \cos \delta, \quad (47)$$

ξ is detection and retention efficiency, T_L, T_R are complete arm power transmissions, ν is task-mode overlap, and δ is differential phase error. A worst-signed additive flag bias b changes the sufficient condition to $g/4 - b > \mu_*$.

A calibrated fiber-coupled superconducting nanowire detector has demonstrated nominal system efficiency 0.98 at 1550 nm [4]. Using this value as an optimistic benchmark, with perfect phase, overlap, masks, and routing, leaves

$$\sqrt{T_L T_R} \nu \cos \delta > 0.922749852200849, \quad (48)$$

corresponding to less than 0.3491602 dB of geometric-mean loss for every non-detector imperfection combined.

A butterfly decomposition of H_{4096} contains twelve stages. One transform on the left path and two on the right give geometric-mean depth eighteen. Under the illustrative assumption of identical stage transmission and no other loss, each stage requires power transmission greater than 0.995543454598465, or loss below 0.0193978 dB.

C. Published scale

Optical Walsh–Hadamard transforms of orders 32 and 64 establish the coherent transform principle in a holographic architecture [8]. Ultrafast time-bin circuits have programmed unitaries through dimension eight and passive networks through 36 modes with reported fidelity above 97 percent [5]. A programmable time-domain Gaussian processor has operated on 216 modes using three loop layers [6]. Complex-medium spatial circuits have used an ambient mixer of approximately 200 modes while characterizing Fourier circuits through dimension seven [7]. A 2026 Gaussian boson-sampling system reports 8,176 optical modes and 4096 connectivity in a different squeezed-state architecture [9].

These experiments establish that large address spaces, time multiplexing, switching, and stable multimode interference are available physical resources. Their reported transformations, states, throughputs, and receivers do not constitute a single-photon H_{4096} implementation satisfying Eq. (46).

D. Evidence assessment

The mathematical result supplies an ideal finite-size active–passive separation and a quantitative implementation target. The reviewed primary evidence does not certify the $N = 4096$ active apparatus at the required end-to-end contrast. This assessment is dated, platform dependent, and separate from Theorem 1.

A device claim would require a measured transfer matrix or task-specific transform certificate, complete arm transmissions, leakage and phase data, source impurity and multiphoton bounds, detector efficiency and dark-count data, a retained rule for no-click outcomes, and a conservative lower confidence bound for $g/4 - b$ above μ_* . Retrying until detection would change the hard-dose experiment and is outside the stated protocol.

IX. DISCUSSION

A. Interpretation

Theorem 1 gives a strict finite-budget separation under one physical interaction meter. A coherent active circuit succeeds with six charged sample traversals, while every classically adaptive number-sector-incoherent fresh-batch passive strategy of charge six fails, despite arbitrary idlers, within-sector quantum states, collective measurements, outcome width, classical computation, and branch-dependent dose allocation.

The mechanism is temporal. The active photon carries phase information from one unknown sign bank through a public transformation and into a second unknown bank before measurement. A passive tree may accumulate arbitrary classical information from previous batches, while

its later quantum amplitudes are prepared fresh. The direct-sum adaptive proof shows that outcome selection changes the correlated law on complete temporal histories without increasing the normalized feature mass or the physical occurrence coefficients.

The finite proof complements the provisional asymptotic $\Omega(N^{1/8})$ passive floor for the broader full fresh-probe model [3]. The asymptotic exponent identifies an unbounded separation, while the $q = 64$ certificate determines that the separation is already strict at $N = 4096$ for the number-sector-incoherent subclass after all physical masks, conditioning costs, coefficient constants, and adaptive quantifiers are retained.

B. Claim boundary

The passive theorem covers classical memory between fresh quantum batches that are block diagonal in total signal photon number. It excludes both coherent quantum memory or coherent control passed between batches and coherence between distinct signal-number sectors within a fresh batch. The latter exclusion is essential to the current certificate: 272 unbalanced high-sector profile-split/state incidences, forming 136 different-number occupation edges, were found during this audit, and their physical coefficients remain unproved. Intermediate models with number-sector coherence, bounded coherent memory, staggered charged interactions inside a nominal round, or restricted inter-batch quantum channels require separate definitions and proofs. Query-adaptivity and quantum-memory separations in other models provide conceptual comparisons [10–12]; their inequalities are not used in the present certificate.

The theorem does not determine the optimal active dose, the optimal passive dose, or the smallest dimension at which a finite separation holds. It does not establish the same passive lower bound when fresh probes have coherence between total signal-number sectors. Active dose five remains open. The existing $N = 1024$ ledger is diagnostic and does not establish a theorem. The signed-permutation hard pair is a proof witness and does not prescribe an experimental input distribution.

C. Experimental boundary

The active resource row is exact at the ideal-circuit level, including the four sign banks, logical mode dimension, path dimension, three photons, two charged traversals per photon, receiver, majority error, and absence of postselection. The scalar robustness condition gives a necessary implementation target within the stated symmetric model. A device-specific error theorem requires measured component data and a noise model that treats loss, leakage, phase, sign errors, dark counts, multiphoton events, and correlations between the three flags.

The current evidence assessment supports a theory-paper statement with explicit experimental requirements. A smaller $N = 1024$ theorem would reduce each sign bank to 1024 coordinates and the total sign count to 4096, but it would still require a dependency-exact outward one-batch certificate below $1/3 - 10^{-3}$ and a device satisfying the same active contrast inequality. That project is distinct from the completed $N = 4096$ proof.

D. Reproducibility

The complete source history is included because the finite certificate depends on a large but finite collection of structural coefficient theorems, independent audits, exact rational bounds, and directed numerical assembly. The manuscript identifies the accepted dependency chain, while the source snapshot preserves rejected routes and counterexamples that prevent superseded inequalities from re-entering the proof through cumulative bookkeeping. The focused and full commands in Sec. X provide the operational record for the theorem statement printed here.

X. VERIFICATION

A. Source package

The paper package contains frozen copies of the complete Round-2, Round-3, and Round-4 source trees under `code/source_snapshot/`. The relative directory names are preserved because the Round-4 suite imports certified functions and runs inherited regressions from the preceding rounds. The full source snapshot contains the physical coefficient searches, independent small-order audits, committed artifacts, registry reconstruction, outward ledger, adaptive theorem tests, active protocol checks, and adversarial tests that rejected earlier incomplete proof routes.

B. Focused verification

The focused command

```
cd code
./run_all.sh
```

runs the paper contract and the theorem-critical Round-4 tests for the active resource row, active robustness threshold, complete outward ledger, adaptive tree frontier, experimental feasibility arithmetic, and paper closeout package. Each artifact is reconstructed in memory and byte-compared with its committed JSON representation when the corresponding source test provides that contract.

C. Complete verification

The command

```
cd code
./run_all.sh --full
```

runs `run_round4_checks.sh` from the frozen snapshot, which first executes every inherited Round-2 and Round-3 regression and then executes the complete Round-4 suite. The accepted run terminates with the registry count 888/888, adaptive frontier multiplier one, passive hard dose six excluded at $N = 4096$, paper package complete, and current-platform experimental credibility not certified.

D. Independent checks

The source history contains explicit counterexamples to superseded inequalities, exact $q = 4$ enumeration, selected direct $q = 8$ signed-permutation sums, arbitrary nonuniform diagonal-law tests, reversal and complement checks, rare-outcome adaptive tests, and independent physical-matrix reconstructions. Passing an inventory or routing regression is not treated as proof of a physical coefficient; only entries with accepted mathematical notes and the appropriate independent validations enter the dependency-exact registry.

Appendix A: The signed-permutation hard pair

1. Coordinate convention

Identify a coordinate in a length- N sign block with a pair $(a, b) \in [q]^2$, using row-major flattening. The normalized Sylvester matrix then factorizes as

$$H_N((a, b), (c, d)) = H_q(a, c)H_q(b, d). \quad (\text{A1})$$

This convention fixes the orientation of the arrays L and R in Eq. (19). Reversal and complement operations used in the coefficient registry preserve the physical simple-support masks under this coordinate identification.

Lemma 2 (Exact plant identity). *For the four blocks in Eq. (21), $F_{4,H}(z) = 1$ for every choice of the three signed permutations. Replacing the first block by its negative gives $F_{4,H}(z) = -1$.*

Proof. Insert the array definitions into Eq. (4) and use $H_N = H_q \otimes H_q$. At the link between $L(P_j)$ and $R(P_j)$, summation over the coordinate not fixed by the signed permutation gives an orthogonality relation of the unnormalized Sylvester matrix,

$$\sum_a S_q(a, r)S_q(a, s) = q \mathbf{1}_{r=s}. \quad (\text{A2})$$

The permutation constraint then identifies the remaining index and the two copies of each sign σ_b cancel. Applying this identity successively to P_1, P_2, P_3 leaves the normalized uniform-state overlap equal to one. Multiplying the first block by -1 changes the four-linear functional by a factor -1 . $\square$

2. Promise concentration

Let F denote the forrelation value of an attenuated positive plant. The exact plant and independent coordinate noise give $\mathbb{E}F = \beta^4$. The accepted concentration argument uses a centered biased-sign subgaussian proxy no larger than one and the four-link chain proxy

$$V_\beta \leq \frac{(1 + \beta^2)(1 + \beta^4)}{N}. \quad (\text{A3})$$

With the promise gap

$$g_\beta = \beta^4 - \frac{1}{4}, \quad (\text{A4})$$

the one-sided Chernoff exponent is bounded below by

$$x_\beta = \frac{Ng_\beta^2}{2(1 + \beta^2)(1 + \beta^4)}. \quad (\text{A5})$$

The numerical certificate avoids an accepted floating evaluation of the exponential. With $m = 4096$,

$$e^{-x_\beta} \leq \left(1 + \frac{x_\beta}{m}\right)^{-m}, \quad (\text{A6})$$

and the right-hand side is evaluated as an exact rational upper. Reflection symmetry gives the same failure bound for the negative plant, and twice the rational upper gives Eq. (28).

Conditioning changes the two transcript distributions by at most the sum of their bad-promise probabilities in total variation. The proof completes the unconditioned adaptive comparison before applying this endpoint inequality.

Appendix B: The dependency-exact ledger

1. Registry counts

The accepted balanced registry is partitioned by proof source as follows.

Status	Entries
Inherited nonuniversal proofs	442
Masked quintic slice	54
Masked local Walsh	180
Masked cubic endpoint	12
Masked double-quintic endpoint	6
Masked double-quintic record	12
Masked four-cubic incidence	38
Masked cubic-septic chain	12
Recovered cubic-quintic endpoint row	28
Joint recovered cubic-quintic	12
Final residual chain	80
Dual-endpoint Schur	12
Total	888

The registry is dependency exact. Before applying the accepted repair theorems, every entry formerly supported only by the invalid universal coefficient-one lemma is reset to its pre-universal state. The actual-mask families are then inserted in dependency order, followed by the final residual theorem and the independently audited dual-endpoint theorem.

The machine-readable artifact records the full 6,016-entry high-sector routing inventory and assigns theorem coefficients only to the 888 number-sector-balanced entries. The remaining 5,128 entries are excluded by Eq. (5); they are not certified coefficient rows. An audit of the broader at-most-six occupation space finds that 272 unbalanced profile splits would contribute split/state incidences, forming 136 undirected edges between different total signal-number sectors. The verifier therefore rejects any unbalanced entry in the theorem coefficient map.

2. Matrix reconstruction

Let $\mathcal{N}_{\leq 6}$ be the set of four-block nonnegative occupation vectors with total at most six. Introducing a slack coordinate for unused dose gives

$$|\mathcal{N}_{\leq 6}| = \binom{6+4}{4} = 210. \quad (\text{B1})$$

The four displayed coordinates are the charged photon occupations in the four sign banks, while the slack coordinate records unused allowance under the hard-dose-six cap and is not a physical mode. Each balanced compatible profile and split contributes a nonnegative weighted edge between states satisfying Eq. (29). The resulting matrix is block diagonal in total signal photon number.

All accepted coefficient values enter through the strict rational grid in Eq. (32). Matrix entries are accumulated with 80-digit decimal arithmetic under upward rounding. When a square root of an integer or rational quantity is required, the stored decimal is advanced until its square is no smaller than the exact argument.

The eigensolver output is used only to construct a positive vector. The accepted spectral bound is the maximum Collatz ratio in Eq. (33). Positivity of the vector and entrywise outward reconstruction of the matrix are checked before the ratio is accepted.

3. Artifact contract

The artifact `q64_complete_outward_ledger.json` records the fixed rational attenuation, the 6,016-split routing inventory, the 888 accepted balanced coefficient rows, the 272-incidence and 136-edge scope audit, rational outward coefficient fields, the 210-state count, the Perron upper, the promise upper, the total upper, the reserve lower, and the pass status. Its deterministic regression rebuilds the complete payload and compares it with the committed file.

The one-batch certificate uses no optimized floating attenuation, no floating eigenvalue as a theorem value, no inward rounding of a square root, no provisional coefficient, and no unbalanced routing placeholder. These restrictions are part of the numerical theorem rather than implementation preferences.

Appendix C: The normalized adaptive factorization

1. Signed transcript functional

Let P_+ and P_- be the unconditioned hard-instance transcript laws of a fixed finite passive tree. Total variation is the supremum of the signed terminal functional over weights $h_\tau \in [-1, 1]$, with the conventional factor $1/2$. Fix one collection of terminal weights. A number-sector-incoherent mixed probe may be spectrally decomposed into pure states, each supported in a fixed total signal-number sector. By convexity, revealing this decomposition label to the controller can only enlarge the class being bounded. Randomized controllers are handled by the same classical refinement.

Index a complete temporal ket history by I and a complete temporal bra history by J . The history includes every batch occupation and all classical choices determined by previous outcomes. Remove the oracle characters from the signed terminal expansion and call the remaining coefficient $D(I, J)$.

2. Induction

At a leaf, $D = h_\tau$ has the scalar factorization

$$h_\tau = \langle v, u \rangle, \quad \|u\|^2 \leq 1, \quad \|v\|^2 \leq 1. \quad (\text{C1})$$

Assume every child subtree selected by a root outcome y has

$$D_y(I, J) = \langle v_{y,J}, u_{y,I} \rangle, \quad (C2)$$

$$\sum_I \|u_{y,I}\|^2 \leq 1, \quad \sum_J \|v_{y,J}\|^2 \leq 1. \quad (C3)$$

Conditional on the refined label, for a pure fixed-number root probe write

$$|\psi\rangle = \sum_i |i\rangle |\alpha_i\rangle, \quad \sum_i \|\alpha_i\|^2 = 1. \quad (C4)$$

Every occupied i in this sum has the same total signal photon number. The complete features take an additional orthogonal direct sum over the refined classical labels.

With root POVM $\{E_y\}_y$, let $a_{y,i} = E_y^{1/2}(|i\rangle |\alpha_i\rangle)$. The complete coefficient is

$$D((i, I), (j, J)) = \sum_y \langle a_{y,j}, a_{y,i} \rangle \langle v_{y,J}, u_{y,I} \rangle. \quad (C5)$$

The features in Eq. (39) reproduce Eq. (C5). Their row mass satisfies

$$\sum_{i,I} \|U_{i,I}\|^2 = \sum_y \left(\sum_i \|a_{y,i}\|^2 \right) \left(\sum_I \|u_{y,I}\|^2 \right) \quad (C6)$$

$$\leq \sum_{y,i} \|a_{y,i}\|^2 = \sum_i \|\alpha_i\|^2 = 1. \quad (C7)$$

The column calculation is identical. This proves Eq. (36) by induction. General outcome spaces follow by finite coarse graining or by the corresponding direct-integral statement.

3. Combination with physical kernels

Orthogonality of the refined labels makes the temporal cross-Gram coefficient vanish unless the ket and bra have equal total signal photon number in every batch. Hence every surviving aggregate pair satisfies $|f(I)| = |f(J)|$ and uses a balanced occurrence split. Let $K_{s,t}$ be one accepted balanced physical occurrence kernel with a Hilbert factorization $K_{s,t} = \langle b_t, c_s \rangle$ and arbitrary-diagonal coefficient $c_{a,t}$. The adaptive expansion pulls this kernel back through aggregate histories as $K_{f(I),f(J)}$. Tensoring the physical factors with the normalized temporal features gives

$$K_{f(I),f(J)} D(I, J) = \langle b_{f(J)} \otimes V_J, c_{f(I)} \otimes U_I \rangle. \quad (C8)$$

The temporal factor therefore acts as one contractive cross-Gram symbol. Its row and column squared masses define two correlated laws of total mass at most one. The

arbitrary-law coefficient scope established by the physical theorems applies directly.

After pushing the two laws to the four-block occupation states, the complete sum is $x^T B y$ with $\|x\|_2, \|y\|_2 \leq 1$. Symmetry and nonnegativity of B give

$$x^T B y \leq \|B\|_{2 \rightarrow 2} = \rho(B). \quad (C9)$$

No common-law assumption and no temporal multiplier are introduced. Conditioning the completed unconditioned laws then gives the total in Eq. (9). If fresh probes have coherence between different total signal-number sectors, the equal-total reduction fails and the unbalanced physical kernels remain an open obligation.

Appendix D: Notation and claim grades

1. Principal notation

Symbol	Meaning
N	logical modes per sign bank, fixed at 4096
q	signed-permutation order, fixed at 64
M	unknown sign coordinates, $4N = 16,384$
H_N	normalized Sylvester transform
D_j	diagonal sign operation for bank j
$F_{4,H}$	four-fold forrelation amplitude
$P_{\text{fresh}}^{\text{ps}}$	number-sector-incoherent classically adaptive fresh-batch passive class
A	active coherent multi-interaction class
D^{hard}	minimum branchwise hard dose at error 1/3
β	attenuation mean, 19/25
P_+, P_-	conditioned positive and negative hard laws
B	nonnegative 210-state occurrence matrix
g	combined retained active contrast

2. Claim grades

Grade	Meaning
Theorem	Analytic proof with every finite numerical input outward certified and included in the dependency registry.
Derived specification	Exact or directed arithmetic derived from an accepted theorem or protocol under an explicitly stated scalar model.
Experimental evidence	Dated comparison with published primary experiments; no role in the mathematical proof.
Diagnostic	Reproducible calculation that does not satisfy every theorem quantifier and is excluded from accepted budgets.

Appendix E: Verification record

1. Result-to-script map

The source programs and regressions in Table II are located in the frozen Round-4 root within the paper package. Except where a directory is printed explicitly, the listed basename occurs once under `searches/` and once under `tests/`.

TABLE II. Focused verification map.

Result	Entry point
Active dose and error	<code>active_six_resource_row.py</code>
Active contrast threshold	<code>active_six_robustness_gate.py</code>
Complete outward ledger	<code>q64_complete_outward_ledger.py</code>
Adaptive multiplier one	<code>q64_adaptive_tree_frontier.py</code>
Feasibility gate	<code>q64_experimental_feasibility_gate.py</code>
Figure data and outputs	<code>figures/generate_figures.py</code> <code>--check</code>
Paper closeout constants	<code>tests/q64_paper_closeout_package.py</code>
Complete inherited suite	<code>run_round4_checks.sh</code>

2. Run of record

The source snapshot was copied from the repository after commit `6aff0c9`, then repaired by the present audit to enforce the number-sector boundary that the original ledger omitted. The focused run reconstructs all 888 balanced entries, rejects any unbalanced routing placeholder in the theorem map, reproduces the 272-incidence and 136-edge scope witness, verifies the classically adaptive frontier multiplier one within the restricted class, and excludes passive hard dose six at $N = 4096$ for that class. The broader fresh-batch claim with signal-number coherence is not certified.

The paper package adds a contract checker that compares manuscript constants and scope language with the audited artifacts, verifies that every conventional citation key resolves in the bibliography and source ledger, and checks the committed figure contract. Separate reviews record mathematical correctness, access-model scope, prose compliance, and the compiled-document read.

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